

Deep Generative Models

Lecture 11

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Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}, \quad \mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

The variance grows since $\sigma(t)$ is a monotonically increasing function.

Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

The variance is preserved if $\mathbf{x}(0)$ has unit variance.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

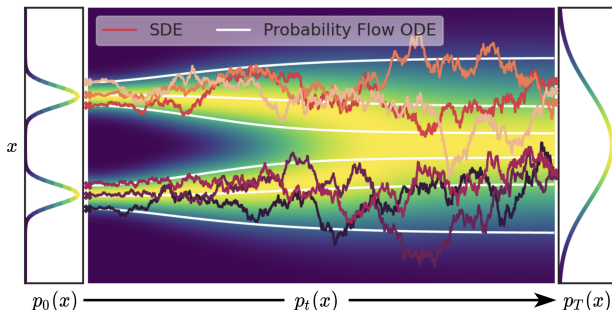
Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad (\text{SDE with the probability path } p_t(\mathbf{x}))$$

Probability Flow ODE

There exists an ODE with the identical probability path $p_t(\mathbf{x})$ of the form:

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Reverse ODE

Let $\tau = 1 - t$ ($d\tau = -dt$):

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1 - \tau)d\tau$$

Reverse SDE

There's a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ in the following form:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t)d\bar{\mathbf{w}}, \quad dt < 0$$

Sketch of the Proof

- ▶ Convert the initial SDE to the probability flow ODE.
- ▶ Reverse the probability flow ODE.
- ▶ Convert the reverse probability flow ODE to the reverse SDE.

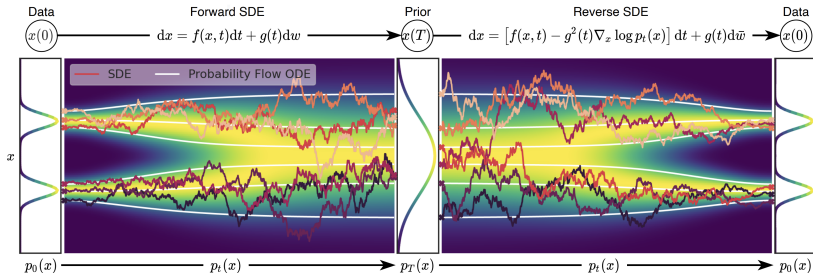
Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t), \quad \text{where} \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad - \text{Probability Flow ODE}$$

$$d\mathbf{x} = \left(\mathbf{v}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t) \right)dt + g(t)d\bar{\mathbf{w}} \quad - \text{Reverse SDE}$$



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Recap of Previous Lecture

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0, 1]} \mathbb{E}_{q(\mathbf{x}(t) | \mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2$$

NCSN

$$q(\mathbf{x}(t) | \mathbf{x}(0)) = \mathcal{N}(\mathbf{x}(0), [\sigma^2(t) - \sigma^2(0)] \cdot \mathbf{I})$$

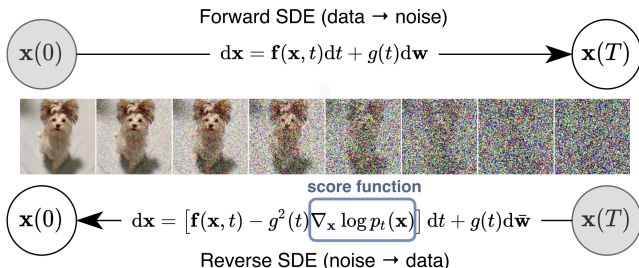
DDPM

$$q(\mathbf{x}(t) | \mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0) e^{-\frac{1}{2} \int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

Recap of Previous Lecture

Sampling

To sample, solve the reverse SDE using numerical solvers (SDEsolve).



- ▶ Discretizing the reverse SDE gives us ancestral sampling.
- ▶ If we use the probability flow ODE instead, then the reverse ODE yields DDIM sampling.

Recap of Previous Lecture

Consider ODE dynamics $\mathbf{x}(t)$ in the interval $t \in [0, 1]$ with $\mathbf{x}_0 \sim p_0(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$, $\mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

Solving the continuity equation ODE numerically is complicated and unstable.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Flow matching is a scalable approach to Neural ODEs.

Outline

1. Conditional Flow Matching (CFM)

- CFM Objective

- Conditional Probability Paths

2. One-Sided Conditioning

Outline

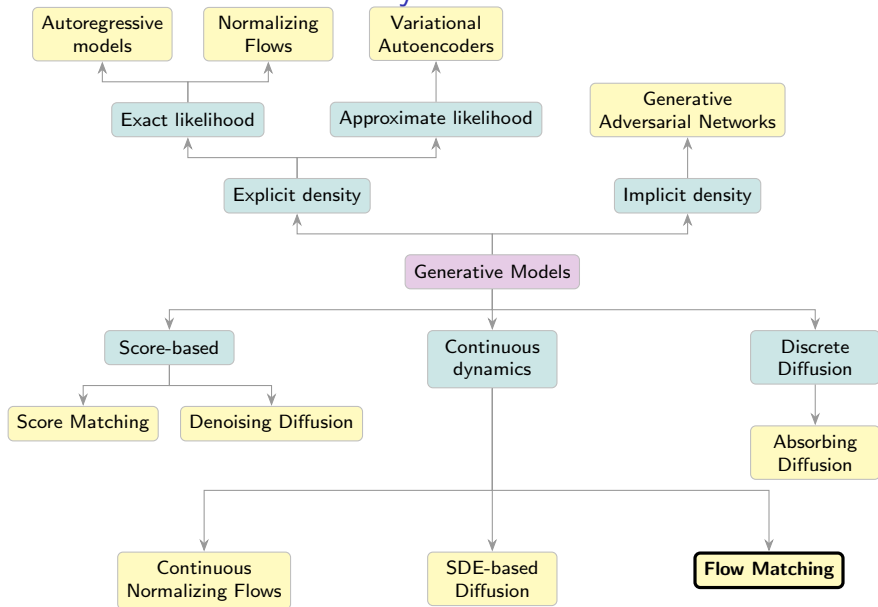
1. Conditional Flow Matching (CFM)

CFM Objective

Conditional Probability Paths

2. One-Sided Conditioning

Generative Models Taxonomy



Outline

1. Conditional Flow Matching (CFM)

CFM Objective

Conditional Probability Paths

2. One-Sided Conditioning

Flow Matching

Let's introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**.

Flow Matching

Let's introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**. The conditional probability path $p_t(\mathbf{x}|\mathbf{z})$ satisfies the KFP theorem:

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

Flow Matching

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where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t) \quad \Rightarrow \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$$

Conditional Vector Field

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

What's the relationship between $\mathbf{v}(\mathbf{x}, t)$ and $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$?

Conditional Vector Field

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What's the relationship between $\mathbf{v}(\mathbf{x}, t)$ and $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$?

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

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Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Conditional Vector Field

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

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Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z} = \int \left(\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} \right) p(\mathbf{z})d\mathbf{z}$$

Conditional Vector Field

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

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Flow Matching

Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, \mathbf{z}, t)\|^2 \rightarrow \min_{\theta}$$

Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

Flow Matching

Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, \mathbf{z}, t)\|^2 \rightarrow \min_{\theta}$$

Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

Proof

This can be proved in a similar way as in the denoising score matching theorem.

Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ &= \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Proof

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ &= \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} [\|\mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] + \text{const}(\theta) \end{aligned}$$

Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Proof

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Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Proof

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} [\|\mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] + \text{const}(\theta) \\ \mathbb{E}_{p_t(\mathbf{x})} [\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] &= \int p_t(\mathbf{x}) \left[\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t) \left(\int p_t(\mathbf{z}|\mathbf{x}) \mathbf{v}(\mathbf{x}, \mathbf{z}, t) d\mathbf{z} \right) \right] d\mathbf{x} = \\ &= \int \int p_t(\mathbf{x}) p_t(\mathbf{z}|\mathbf{x}) [\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, \mathbf{z}, t)] d\mathbf{z} d\mathbf{x} \end{aligned}$$

Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Proof

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} [\|\mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] + \text{const}(\theta) \\ \mathbb{E}_{p_t(\mathbf{x})} [\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] &= \int p_t(\mathbf{x}) \left[\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t) \left(\int p_t(\mathbf{z}|\mathbf{x}) \mathbf{v}(\mathbf{x}, \mathbf{z}, t) d\mathbf{z} \right) \right] d\mathbf{x} = \\ &= \int \int p_t(\mathbf{x}) p_t(\mathbf{z}|\mathbf{x}) [\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, \mathbf{z}, t)] d\mathbf{z} d\mathbf{x} = \\ &= \mathbb{E}_{p(\mathbf{z})} \mathbb{E}_{p_t(\mathbf{x}|\mathbf{z})} [\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, \mathbf{z}, t)] \end{aligned}$$

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

Training

1. Sample $t \sim U[0, 1]$, $\mathbf{z} \sim p(\mathbf{z})$, $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
2. Compute loss $\mathcal{L} = \|\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

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Training

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2. Compute loss $\mathcal{L} = \|\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

Marginal-to-Conditional: Scores vs Velocities

Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma | \mathbf{x})} \|\mathbf{s}_{\theta, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma | \mathbf{x})\|_2^2 \rightarrow \min_{\theta}$$

Optimal predictor:

$$\mathbf{s}_{\theta^*, \sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x} | \mathbf{x}_\sigma)} [\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma | \mathbf{x})] = \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$$

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In both cases, the optimal MSE predictor is the posterior mean of the **conditional** target, which equals the intractable **marginal** quantity (score or velocity).

Outline

1. Conditional Flow Matching (CFM)

CFM Objective

Conditional Probability Paths

2. One-Sided Conditioning

Conditional Flow Matching: Continuity Equations

Continuity Equations

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t) \cdot p_t(\mathbf{x}))$$
$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) \cdot p_t(\mathbf{x}|\mathbf{z}))$$

What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

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- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.

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- ▶ $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- ▶ But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t)p_t(\mathbf{x})) = 0$ can generate a valid probability path.

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- ▶ But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t)p_t(\mathbf{x})) = 0$ can generate a valid probability path.
- ▶ Moreover, not every given path $p_t(\mathbf{x})$ can actually arise from particles moving under some velocity field $\mathbf{v}(\mathbf{x}, t)$.

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

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Vector fields

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

Conditional Flow Matching: Paths and Vector Fields

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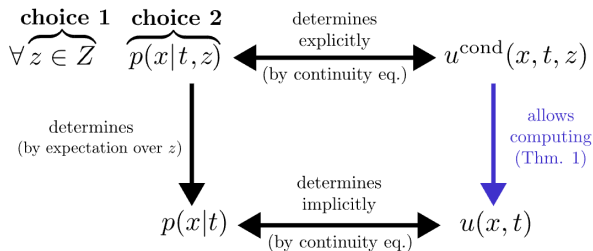
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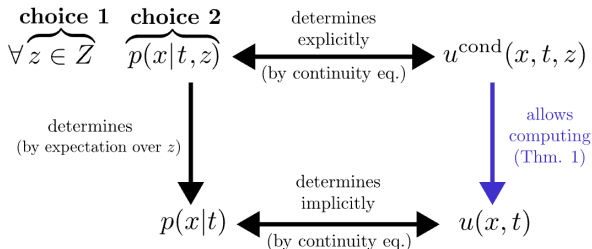
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- ▶ $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ and $p_t(\mathbf{x}|\mathbf{z})$ uniquely determine $\mathbf{v}(\mathbf{x}, t)$.
- ▶ But marginal velocity does not determine conditional velocities (many decompositions exist).

From Marginal to Conditional Paths



From Marginal to Conditional Paths



Open Questions

- Q1** How should we choose the convenient conditioning latent variable $\mathbf{z}$?
- Q2** How can we parametrize $p_t(\mathbf{x})$ (or $p_t(\mathbf{x}|\mathbf{z})$) to enforce the following constraints?

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

Gaussian Conditional Probability Paths [Q2]

Let's consider the following parametrization:

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z}))$$

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The concrete $\boldsymbol{\mu}_t(\mathbf{z})$, $\boldsymbol{\sigma}_t(\mathbf{z})$ (which satisfies the boundary constraints) will be fixed once we choose $\mathbf{z}$ [Q1].

Gaussian flow

- ▶ Let's derive the vector field for **any** Gaussian path $p_t(\mathbf{x}|\mathbf{z})$.
- ▶ There are infinitely many flows $\psi_t(\mathbf{x}_0, \mathbf{z})$ (or, equivalently, vector fields $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$) that generate a particular probability path $p_t(\mathbf{x}|\mathbf{z})$.

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- ▶ Let's consider the following flow:

$$\mathbf{x}_t = \psi_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon}$ is fixed by the condition: $\mathbf{x}_0 = \boldsymbol{\mu}_0(\mathbf{z}) + \boldsymbol{\sigma}_0(\mathbf{z}) \odot \boldsymbol{\epsilon}$.

Gaussian Conditional Vector Field

Gaussian Conditional Probability Path

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How to derive the expression for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$ for this particular flow $\psi_t(\mathbf{x}_0, \mathbf{z})$?

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$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t); \quad \boldsymbol{\epsilon} = \frac{1}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

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Endpoint Conditioning

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Let define our latent variable $\mathbf{z}$.

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Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = \mathbf{x}_1$. Then $p(\mathbf{z}) = p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_1)p_1(\mathbf{x}_1)d\mathbf{x}_1$$

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Gaussian Conditional Probability Path

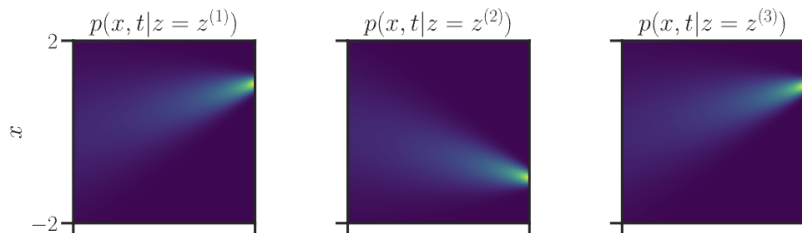
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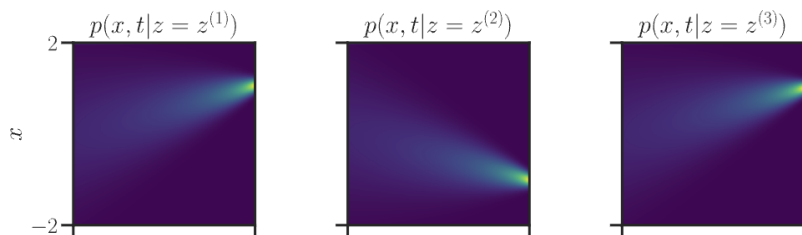


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Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases}$$

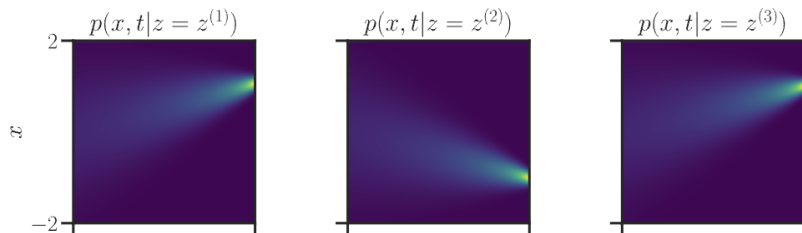
image credit: *A Visual Dive into Conditional Flow Matching*

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image credit: A Visual Dive into Conditional Flow Matching

One-Sided Conditioning: Conditional Vector Field

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Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mu'_t(\mathbf{x}_1) + \frac{\sigma'_t(\mathbf{x}_1)}{\sigma_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \mu_t(\mathbf{x}_1))$$

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$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mu'_t(\mathbf{x}_1) + \frac{\sigma'_t(\mathbf{x}_1)}{\sigma_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \mu_t(\mathbf{x}_1))$$

$$\begin{aligned}\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) &= \mathbf{x}_1 - \frac{1}{1-t} \cdot (\mathbf{x}_t - t\mathbf{x}_1) = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1-t} = \\ &= \frac{\mathbf{x}_1 - t\mathbf{x}_1 - (1-t)\mathbf{x}_0}{1-t}\end{aligned}$$

One-Sided Conditioning: Conditional Vector Field

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

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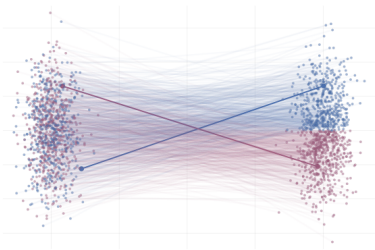
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The conditional vector field $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t)$ defines straight lines between $p_{\text{data}}(\mathbf{x})$ and $\mathcal{N}(0, \mathbf{I})$.

One-Sided Conditioning: CFM Objective

$$\begin{aligned} & \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{z \sim p(z)} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|z)} \left\| \mathbf{v}(\mathbf{x}, z, t) - \mathbf{v}_\theta(\mathbf{x}, t) \right\|^2 = \\ & = \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_1)} \left\| \frac{\mathbf{x}_1 - \mathbf{x}}{1 - t} - \mathbf{v}_\theta(\mathbf{x}, t) \right\|^2 \end{aligned}$$

One-Sided Conditioning: CFM Objective

$$\begin{aligned} & \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_1)} \left\| \frac{\mathbf{x}_1 - \mathbf{x}}{1 - t} - \mathbf{v}_\theta(\mathbf{x}, t) \right\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(t\mathbf{x}_1 + (1 - t)\mathbf{x}_0, t)\|^2 \end{aligned}$$

One-Sided Conditioning: CFM Objective

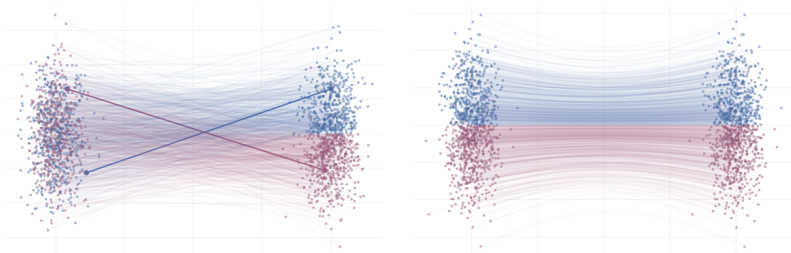
$$\begin{aligned} & \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_1)} \left\| \frac{\mathbf{x}_1 - \mathbf{x}}{1 - t} - \mathbf{v}_\theta(\mathbf{x}, t) \right\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(t\mathbf{x}_1 + (1 - t)\mathbf{x}_0, t)\|^2 \end{aligned}$$

- ▶ We fit straight lines between the noise distribution $p(\mathbf{x})$ and the data distribution $p_{\text{data}}(\mathbf{x})$.
- ▶ The **marginal** path $p_t(\mathbf{x})$ does not give straight lines.

One-Sided Conditioning: CFM Objective

$$\begin{aligned} & \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_1)} \left\| \frac{\mathbf{x}_1 - \mathbf{x}}{1-t} - \mathbf{v}_\theta(\mathbf{x}, t) \right\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, t)\|^2 \end{aligned}$$

- ▶ We fit straight lines between the noise distribution $p(\mathbf{x})$ and the data distribution $p_{\text{data}}(\mathbf{x})$.
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CFM with One-Sided Conditioning: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\theta}(\mathbf{x}_t, t)\|^2 \rightarrow \min_{\theta}$$

Training

1. Sample $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$, $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$, $t \sim U[0, 1]$.
2. Compute noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\theta}(\mathbf{x}_t, t)\|^2$.

CFM with One-Sided Conditioning: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\theta}(\mathbf{x}_t, t)\|^2 \rightarrow \min_{\theta}$$

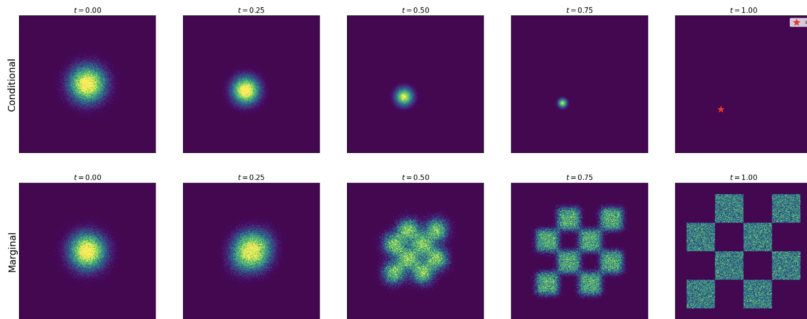
Training

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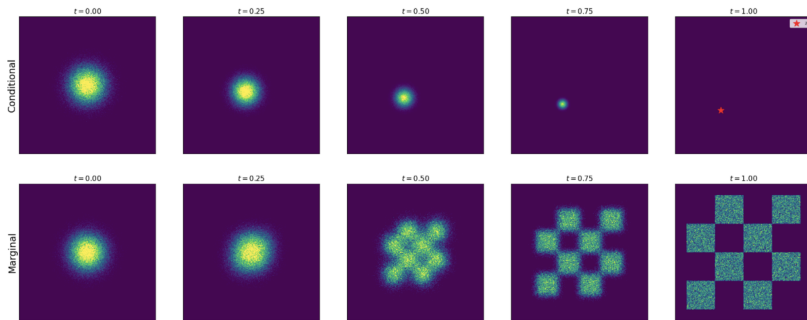
Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:
$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

Conditional vs. Marginal Paths



Conditional vs. Marginal Paths



- ▶ One-sided conditioning gives us the way to construct generative model.
- ▶ Now we extend it to image-to-image formulation (mapping between two distinct $p_{\text{data}_1}(\mathbf{x})$ and $p_{\text{data}_2}(\mathbf{x})$).

Summary

- ▶ Conditional flow matching introduces the latent variable $\mathbf{z}$, reformulating the initial task in terms of conditional dynamics.
- ▶ The marginal vector field can be expressed as the posterior mean of conditional vector fields: $\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$.
- ▶ The FM and CFM objectives have the same optimal solution, making CFM a tractable alternative to FM.
- ▶ For Gaussian conditional paths $p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z}))$, the conditional vector field has a closed form for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$.
- ▶ One-sided conditioning uses endpoint conditioning $\mathbf{z} = \mathbf{x}_1$ with straight conditional paths, yielding the simple vector field $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mathbf{x}_1 - \mathbf{x}_0$.